Learning Expressive Humanoid Locomotion from Monocular Runway Videos

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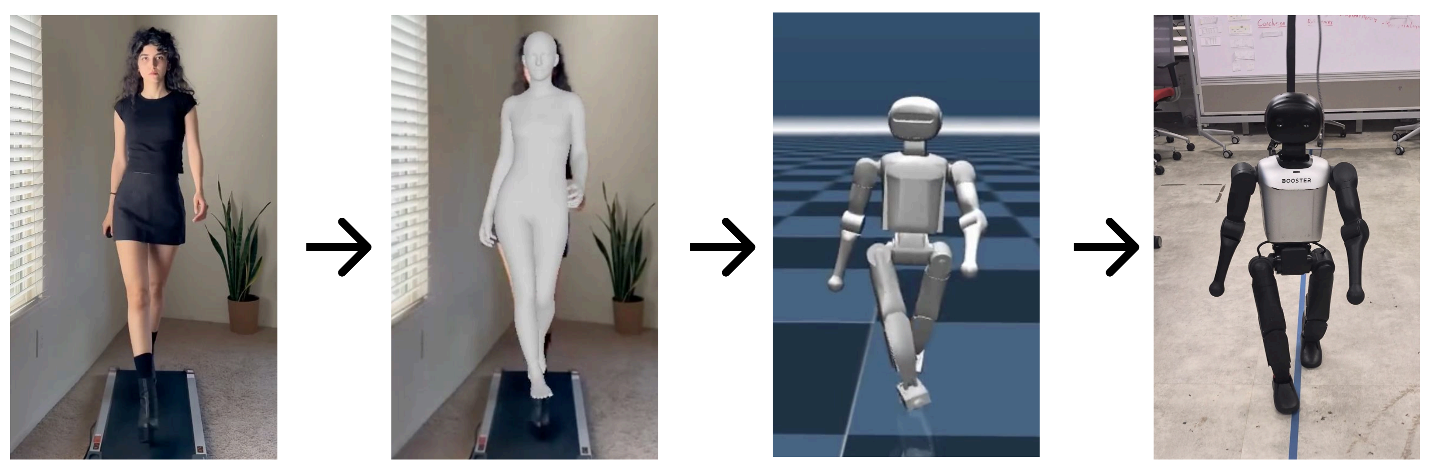
code + pipeline

Why runway gait?

Runway walking uses posture, stride, and foot placement to present a garment. Humanoids at fashion shows run locomotion policies tuned for balance and velocity tracking — they cross the runway, but they do not perform it. We learn the gait directly from a single video.



(a) Monocular runway video → (b) Recovered 3D human motion → (c) Retargeted motion in simulation → (d) Physical Booster K1

Prior work on fashion-inspired gait stayed in simulation or in planar models.

We deploy on physical hardware

From video to a running policy

monocular video
↓
GVHMR — 3D human motion
↓
GMR — retarget to K1
↓
ground correction
↓
BeyondMimic — tracking policy
↓
MuJoCo — sim-to-sim
↓
Booster K1 hardware

Ground correction
Motion snapped to the floor; drift removed without changing joint angles.

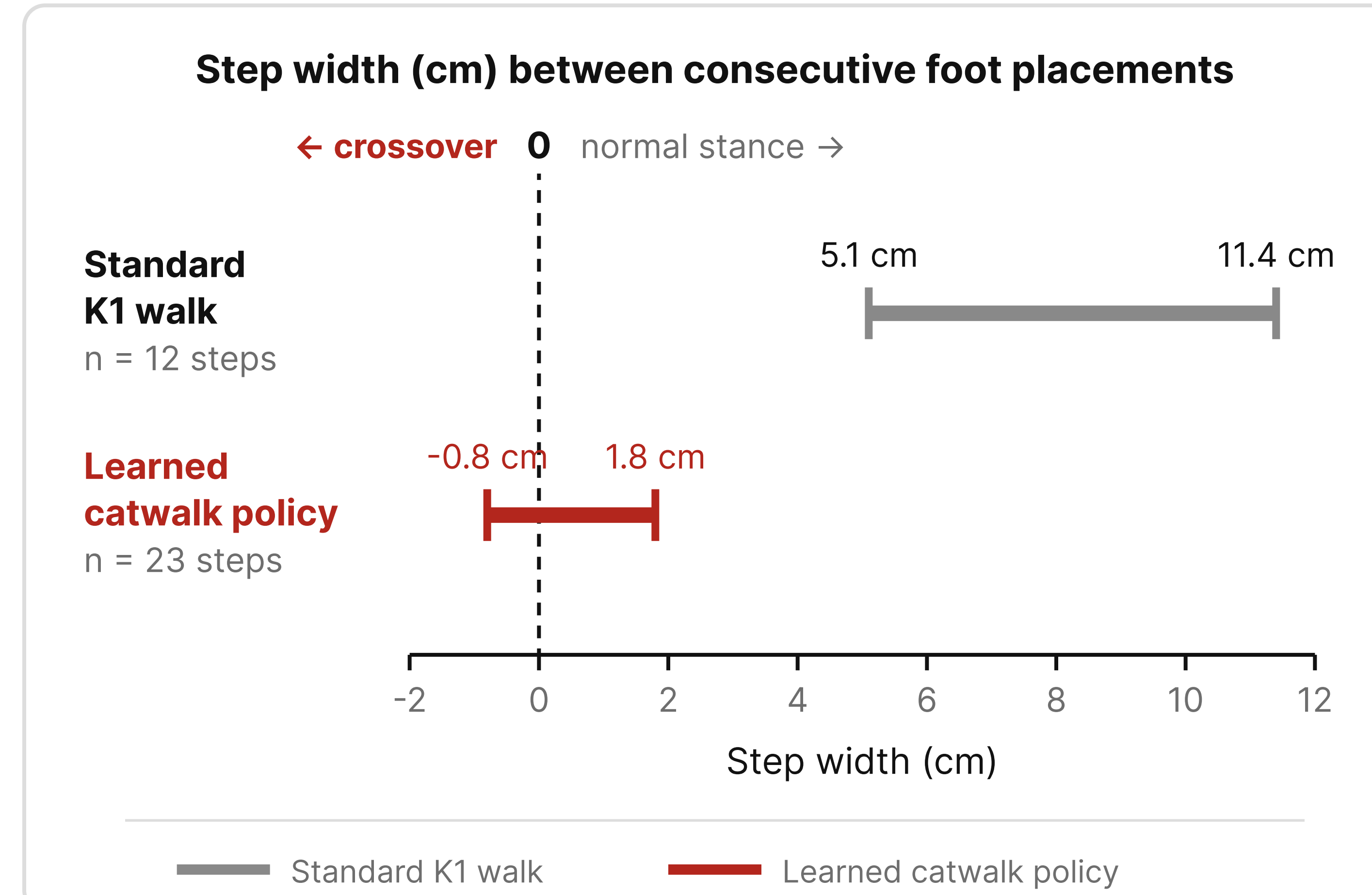
Consistent settings
Actuator limits, stiffness and damping identical in training and on hardware.

Why step width?
Kobayashi et al. (2024) found leg crossing to be what separates runway walking from ordinary gait, and step width measures it directly.

Results in the real world

20/20

physical trials completed without a fall ≈ 23 steps each



Standard walk	5.1 – 11.4 cm	
Catwalk policy	−0.8 – 1.8 cm	(crossover steps present)

Watch the robot walk
Loop of one full runway traversal.

Future work

- Controllable policy for speed and direction.
- Additional quantitative metrics beyond step width.
- Expert evaluation of walking style.

References

GVHMR (2024) · GMR (2025) · BeyondMimic (2025)
Kobayashi et al. (2024) · Or (2012) · Huzaifa et al. (2020)

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